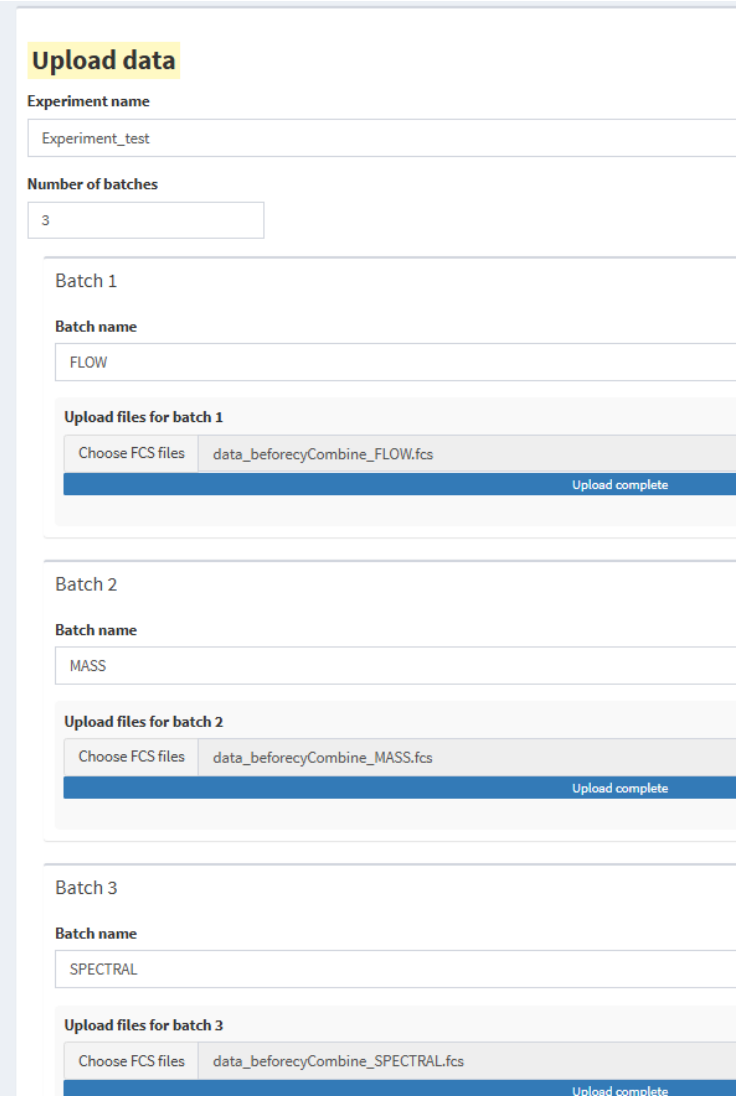
cyCombine app tutorial

Implementation of the existing cyCombine tool as a Shiny interface (PMID: 35361793).

1) Upload data



Data Management:

a) Enter the number of batches
b) Upload FCS files for each batch

2) Panel Harmonization

Panel Harmonization

Export panel (Excel)

Import modified panel

Browse... No file selected

	FLOW_Marker	MASS_Fluo	MASS_Marker	SPECTRAL_Fluo	SPECTRAL_Marker	Normalize
$P1N	CD4__CoreMarkerA_	CD4__CoreMarkerA_	CD4__CoreMarkerA_	CD4__CoreMarkerA_	CD4__CoreMarkerA_	☐
$P2N	CD11b__CoreMarkerB_	CD11b__CoreMarkerB_	CD11b__CoreMarkerB_	CD11b__CoreMarkerB_	CD11b__CoreMarkerB_	☐
$P3N	CD44__CoreMarkerC_	CD44__CoreMarkerC_	CD44__CoreMarkerC_	CD44__CoreMarkerC_	CD44__CoreMarkerC_	☐
$P4N	CD5/Ly6G__CoreMarkerD_	CD5/Ly6G__CoreMarkerD_	CD5/Ly6G__CoreMarkerD_	CD5/Ly6G__CoreMarkerD_	CD5/Ly6G__CoreMarkerD_	☐
$P5N	IA/IE__CoreMarkerE_	IA/IE__CoreMarkerE_	IA/IE__CoreMarkerE_	IA/IE__CoreMarkerE_	IA/IE__CoreMarkerE_	☐
$P6N	CD317__CoreMarkerF_	CD317__CoreMarkerF_	CD317__CoreMarkerF_	CD317__CoreMarkerF_	CD317__CoreMarkerF_	☐
$P7N	CD62L__CoreMarkerG_	CD62L__CoreMarkerG_	CD62L__CoreMarkerG_	CD62L__CoreMarkerG_	CD62L__CoreMarkerG_	☐
$P8N	CD8a/SiglecF__CoreMarkerH_	CD8a/SiglecF__CoreMarkerH_	CD8a/SiglecF__CoreMarkerH_	CD8a/SiglecF__CoreMarkerH_	CD8a/SiglecF__CoreMarkerH_	☐
$P9N	F4/80__CoreMarkerI_	F4/80__CoreMarkerI_	F4/80__CoreMarkerI_	F4/80__CoreMarkerI_	F4/80__CoreMarkerI_	☐
$P10N	CD161__CoreMarkerJ_	CD161__CoreMarkerJ_	CD161__CoreMarkerJ_	CD161__CoreMarkerJ_	CD161__CoreMarkerJ_	☐
$P11N	Ly6C__CoreMarkerK_	Ly6C__CoreMarkerK_	Ly6C__CoreMarkerK_	Ly6C__CoreMarkerK_	Ly6C__CoreMarkerK_	☐
$P12N	CD25__CoreMarkerL_	CD25__CoreMarkerL_	CD25__CoreMarkerL_	CD25__CoreMarkerL_	CD25__CoreMarkerL_	☐
$P13N	CD11c__CoreMarkerM_	CD11c__CoreMarkerM_	CD11c__CoreMarkerM_	CD11c__CoreMarkerM_	CD11c__CoreMarkerM_	☐
$P14N	CD19__CoreMarkerN_	CD19__CoreMarkerN_	CD19__CoreMarkerN_	CD19__CoreMarkerN_	CD19__CoreMarkerN_	☐
$P15N	afterCyCombine_UMAP1	afterCyCombine_UMAP1	afterCyCombine_UMAP1	afterCyCombine_UMAP1	afterCyCombine_UMAP1	☐

Please click the 'Validate Design' button before proceeding

There is a covariable ? *In cyCombine, a covariate is a biological or experimental variable within a batch that should be preserved during batch correction.

No

Validate Design

This section allows you to match marker names across files from different batches.

Two recommended workflows depending on the situation:

Case 1: Identical panels across batches (markers written in the same way)

Case 2: Different panels

2) Panel Harmonization
Case 1: Identical panels across batches (markers written in the same way)

Panel Harmonization

Export panel (Excel)

Import modified panel

Browse... No file selected

	Batch1_Fluo	Batch1_Marker	Batch2_Fluo	Batch2_Marker	Normalize
$P25N	Nd142Di	CD11c	Nd142Di	CD11c	☐
$P26N	Nd143Di	CD357	Nd143Di	CD357	☐
$P27N	Nd144Di	144Nd	Nd144Di	144Nd	☐
$P28N	Nd145Di	CD4	Nd145Di	CD4	☑
$P29N	Nd146Di	CD5	Nd146Di	CD5	☑
$P30N	Sm147Di	147Sm	Sm147Di	147Sm	☐
$P31N	Nd148Di	CD11b	Nd148Di	CD11b	☑
$P32N	Sm149Di	149Sm	Sm149Di	149Sm	☐
$P33N	Nd150Di	150Nd	Nd150Di	150Nd	☐
$P34N	Eu151Di	CD25	Eu151Di	CD25	☑
$P35N	Sm152Di	152Sm	Sm152Di	152Sm	☐
$P36N	Eu153Di	153Eu	Eu153Di	153Eu	☐
$P37N	Sm154Di	siglecF	Sm154Di	siglecF	☑
$P38N	Gd155Di	155Gd	Gd155Di	155Gd	☐
$P39N	Gd156Di	XCR1	Gd156Di	XCR1	☐
$P40N	Gd157Di	157Gd	Gd157Di	157Gd	☐

validate design

Case 1: Identical panels across batches (markers written in the same way)

In this case, simply select the markers to normalize in the table.

Then click **"Validate Design."**

2) Panel Harmonization
Case 2: Different panels

Panel Harmonization

Export panel (Excel)

Import modified panel

Browse... No file selected

	Batch1_Fluo	Batch1_Marker	Batch2_Fluo	Batch2_Marker	Normalize
$P1N	Time	Time	Time	Time	☐
$P2N	Event_length	length	Event_length	length	☐
$P3N	Y89Di	CD45	Y89Di	CD45	☐
$P4N	Pd102Di	102Pd	Pd102Di	102Pd	☐
$P5N	Rh103Di	103Rh	Rh103Di	103Rh	☐
$P6N	Pd104Di	104Pd	Pd104Di	104Pd	☐
$P7N	Pd105Di	105Pd	Pd105Di	105Pd	☐
$P8N	Pd106Di	106Pd	Pd106Di	106Pd	☐
$P9N	Pd108Di	108Pd	Pd108Di	108Pd	☐
$P10N	Pd110Di	110Pd	Pd110Di	110Pd	☐
$P11N	Cd111Di	111Cd	Cd111Di	111Cd	☐
$P12N	Cd112Di	112Cd	Cd112Di	112Cd	☐
$P13N	Cd113Di	113Cd	Cd113Di	113Cd	☐
$P14N	Cd114Di	114Cd	Cd114Di	114Cd	☐
$P15N	In115Di	115In	In115Di	115In	☐
$P16N	Cd116Di	116Cd	Cd116Di	116Cd	☐

validate design

Case 2: Different panels
Recommended workflow:
1) Export the panel

2) Panel Harmonization
Case 2: Different panels

	A	B	C	D	E	F
1	Batch1_Fluo	Batch1_Mark	Batch2_Fluo	Batch2_Mark	Normalize	
2	Nd142Di	CD11c	R-Alexa700-A	CD11c	FALSE	
3	Nd145Di	CD4	UV-BUV395-A	CD4	FALSE	
4	Nd148Di	CD11b	UV-BUV661-A	CD11b	FALSE	
5	Eu151Di	CD25	R-APC-A	CD25	FALSE	
6	Gd160Di	CD62L	V-BV711-A	CD62L	FALSE	
7	Er166Di	CD19	R APC-Cy7-A	CD19	FALSE	
8	Er170Di	CD161	G-PE-Cy5-A	CD161	FALSE	
9	Yb171Di	CD44	UV-BUV737-A	CD44	FALSE	
10	Yb173Di	CD317	V-BV650-A	CD317	FALSE	
11	Time	Time			FALSE	
12	Event_length	length			FALSE	
13	Y89Di	CD45			FALSE	
14	Pd102Di	102Pd			FALSE	
15	Rh103Di	103Rh			FALSE	
16	Pd104Di	104Pd			FALSE	
17	Pd105Di	105Pd			FALSE	
18	Pd106Di	106Pd			FALSE	
19	Pd108Di	108Pd			FALSE	
20	Pd110Di	110Pd			FALSE	
21	Cd111Di	111Cd			FALSE	
22	Cd112Di	112Cd			FALSE	
23	Cd113Di	113Cd			FALSE	
24	Cd114Di	114Cd			FALSE	
25	In115Di	115In			FALSE	
26	Cd116Di	116Cd			FALSE	
27	Sn120Di	120Sn			FALSE	
28	I127Di	127I			FALSE	
29	Xe131Di	131Xe			FALSE	
30	Cs133Di	133Cs			FALSE	
31	Ba138Di	138Ba			FALSE	
32	La139Di	139La			FALSE	
33	Ce140Di	140Ce			FALSE	
34	Pr141Di	Ly6G			FALSE	
35	Nd143Di	CD357			FALSE	
36	Nd144Di	144Nd			FALSE	

Panel

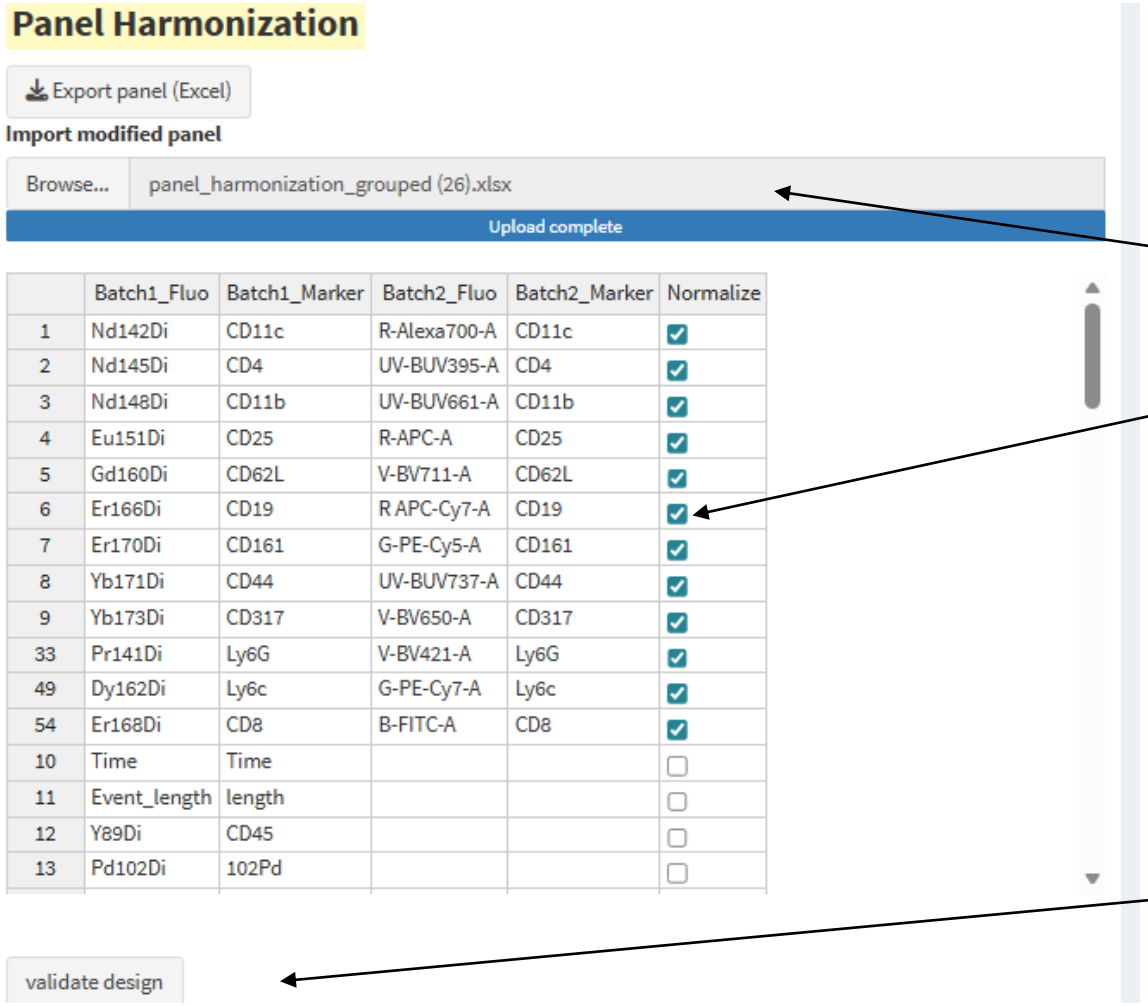
Prêt

In the exported Excel file, the markers highlighted in **green** are those that match across your different batches. Those in **red** are the ones that do not match.

Modify the **Marker** columns as shown in the video (**do not change the Fluo columns, only the marker names**).
You do not need to modify all marker names—only those you want to normalize.

Once finished, save the table.

2) Panel Harmonization
Case 2: Different panels



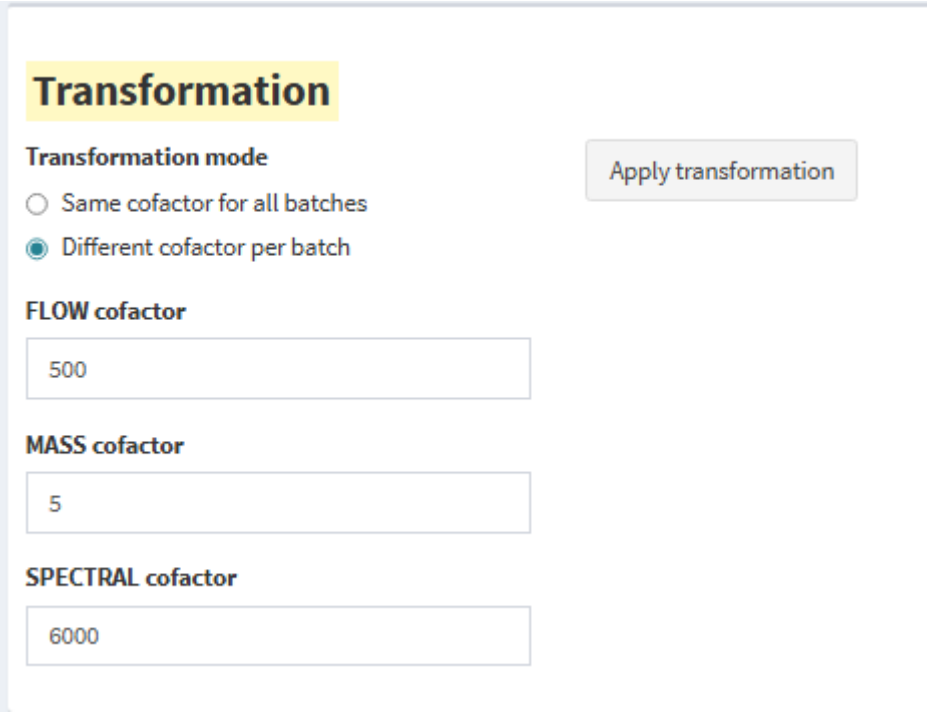
	Batch1_Fluo	Batch1_Marker	Batch2_Fluo	Batch2_Marker	Normalize
1	Nd142Di	CD11c	R-Alexa700-A	CD11c	☑
2	Nd145Di	CD4	UV-BUV395-A	CD4	☑
3	Nd148Di	CD11b	UV-BUV661-A	CD11b	☑
4	Eu151Di	CD25	R-APC-A	CD25	☑
5	Gd160Di	CD62L	V-BV711-A	CD62L	☑
6	Er166Di	CD19	R APC-Cy7-A	CD19	☑
7	Er170Di	CD161	G-PE-Cy5-A	CD161	☑
8	Yb171Di	CD44	UV-BUV737-A	CD44	☑
9	Yb173Di	CD317	V-BV650-A	CD317	☑
33	Pr141Di	Ly6G	V-BV421-A	Ly6G	☑
49	Dy162Di	Ly6c	G-PE-Cy7-A	Ly6c	☑
54	Er168Di	CD8	B-FITC-A	CD8	☑
10	Time	Time			☐
11	Event_length	length			☐
12	Y89Di	CD45			☐
13	Pd102Di	102Pd			☐

Re-import the modified panel into the application.

Make sure to **check the markers to normalize** if you haven't done so already.

When finished, click **"Validate Design"**.

3) Transformation

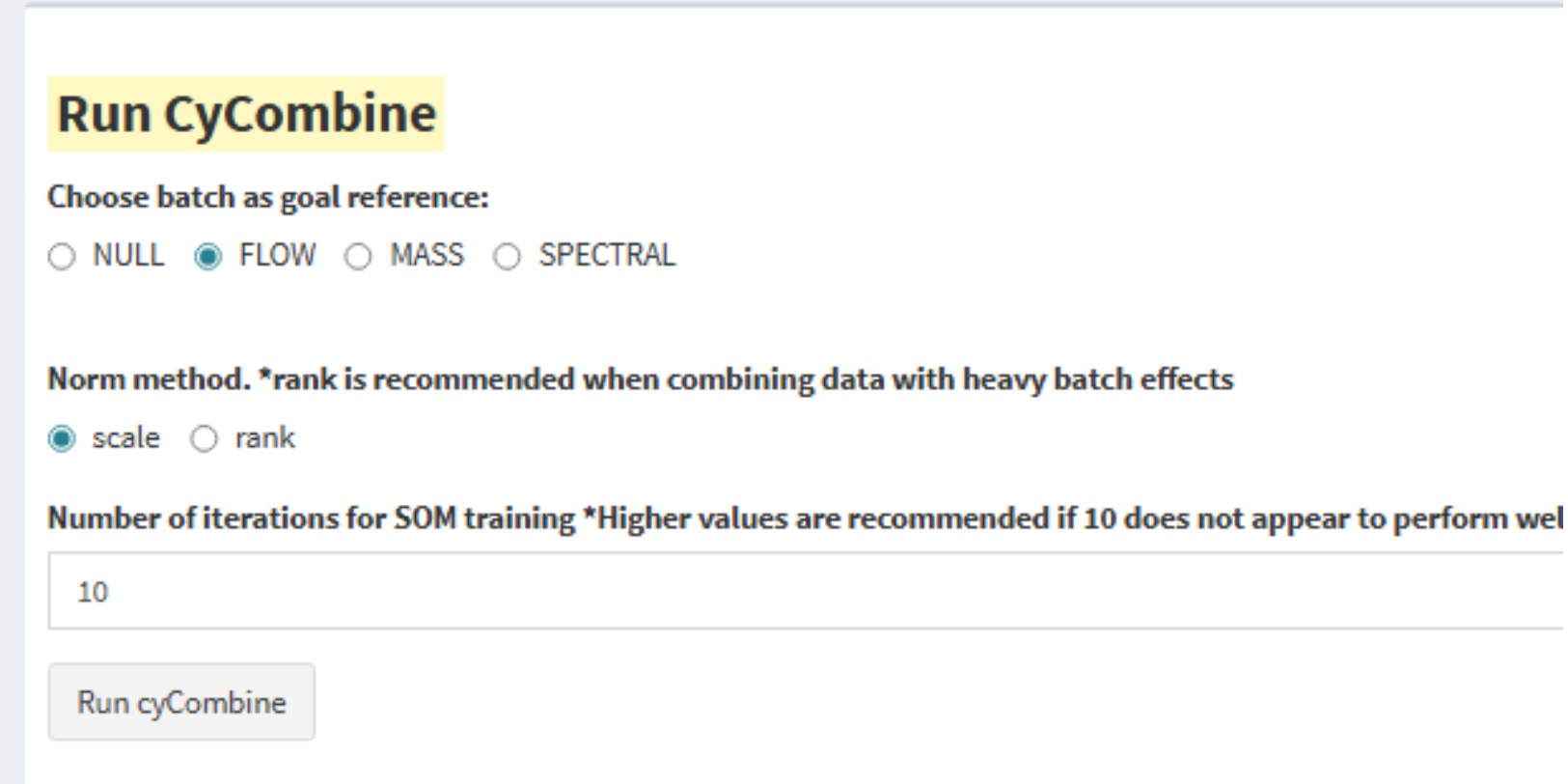


The files are compensated upon import into the application if they contain a compensation matrix.
Regarding the transformation, since **cyCombine** can integrate datasets from different techniques, it is necessary to be able to transform your data using an arcsinh transformation with a different cofactor for each batch according to the technique.

Recommended cofactors:

Mass: 5
Flow: 500
Spectral: 6000

4) Run cyCombine



Reference batch: You can use a batch as a reference batch, which will allow your other batches to align with it

.

Norm method: Leave as *scale* if using the same techniques (e.g., Flow/Flow); otherwise, use *rank* (e.g., Flow/Mass).

5) Visualization
a) Histogram

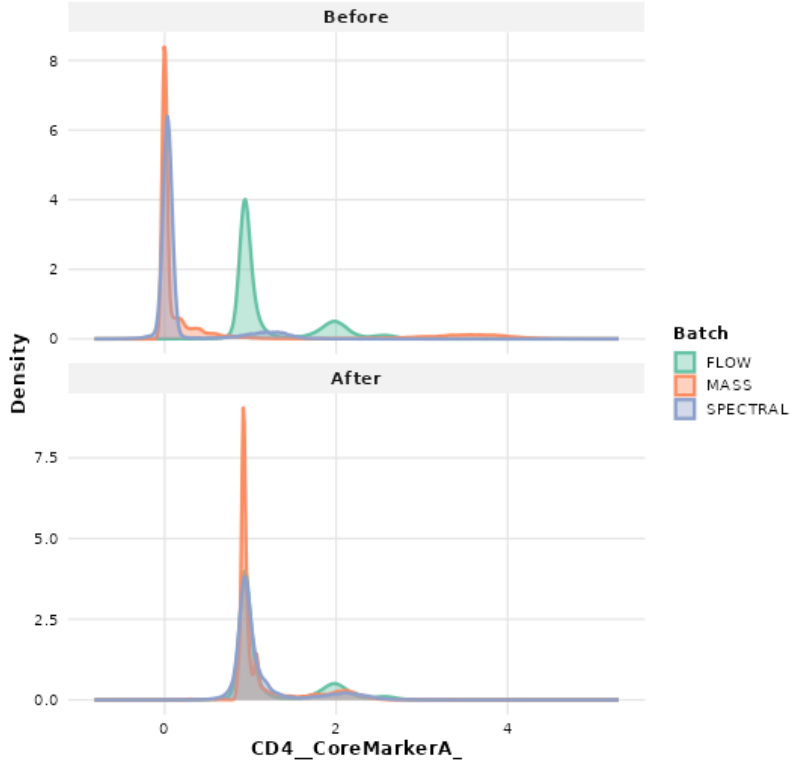
Visualization

Histogram

Select marker

CD4__CoreMarkerA_

ırker Density Before vs After cyCombine: CD4__CoreMarkerA_



5) Visualization
b) ScatterPlot density

File for FLOW

data_beforecyCombine_FLOW.fcs

File for MASS

data_beforecyCombine_MASS.fcs

File for SPECTRAL

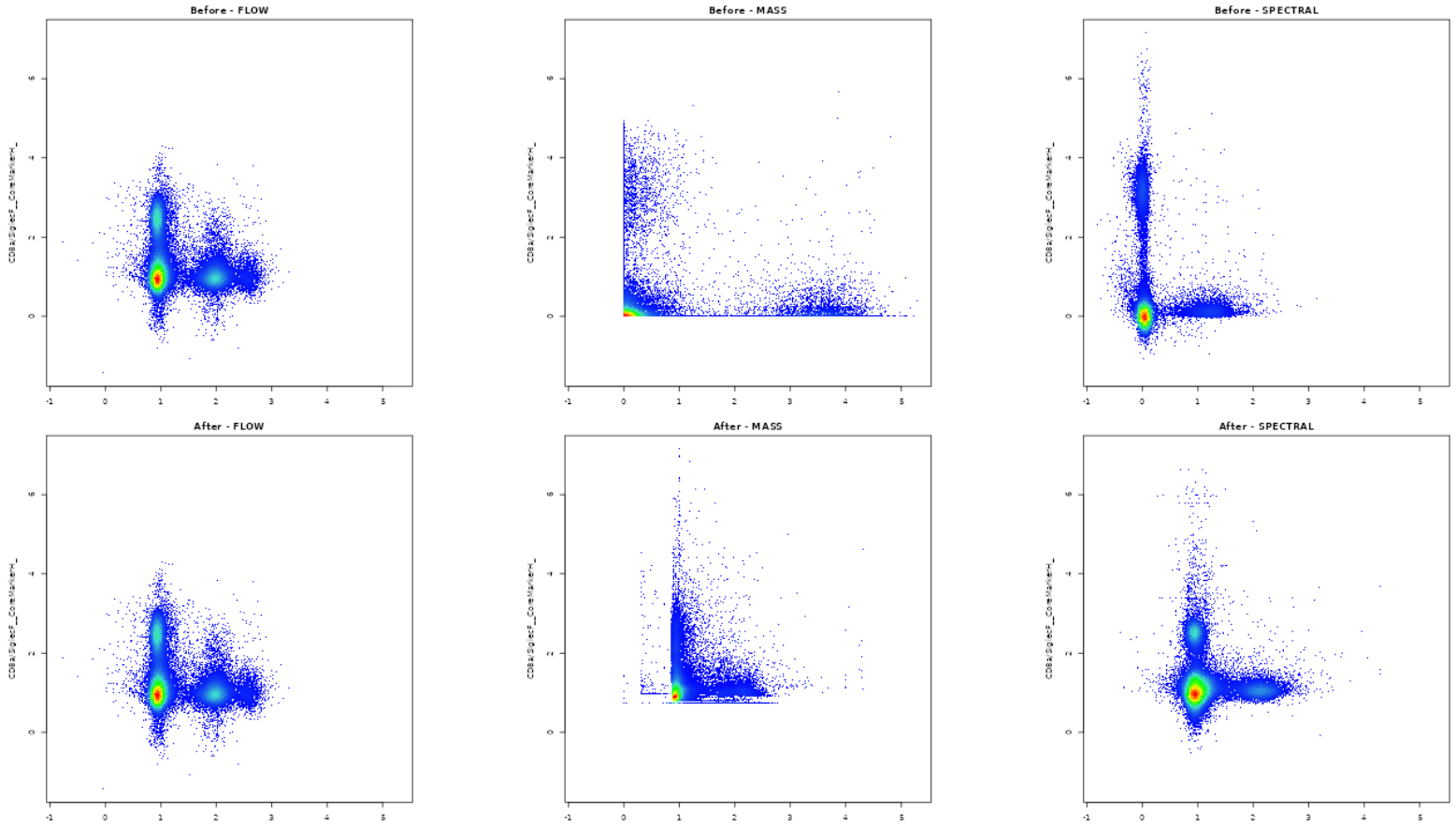
data_beforecyCombine_SPECTRAL.fcs

Marker X

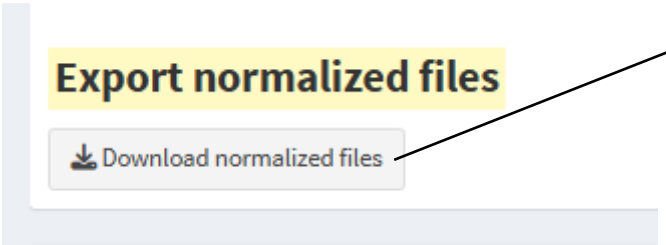
CD4__CoreMarkerA_

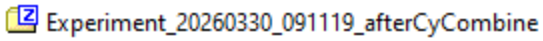
Marker Y

CD8a/SiglecF__CoreMarkerH_



6) Export





Output: You will receive a ZIP file containing your files after cyCombine, **un-compensated and de-transformed**.